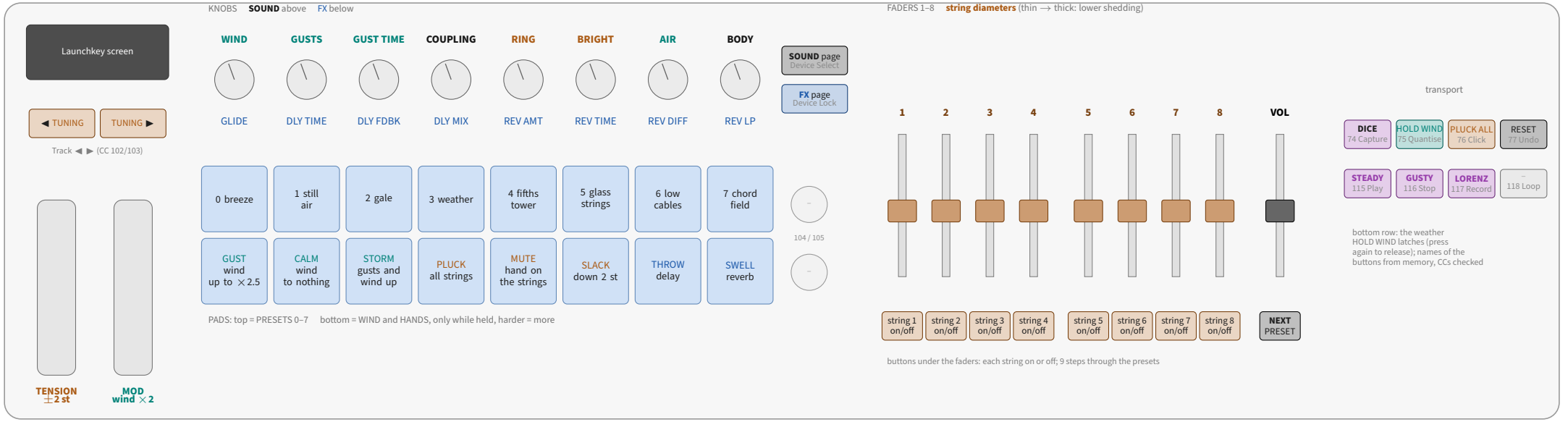
aeolian

Axoloti Core + Launchkey 49 MK3 (standalone mode) · a wind harp: eight strings, the wind plays them · `patches/aeolian/aeolian_v1.axp`



How it sounds

Eight strings in the wind. Wind across a string sheds vortices at the Strouhal frequency $f = 0.2\,U/d$ (wind U, diameter d). When f comes near one of the string's harmonics, the wake *locks in* and that harmonic sings; as the wind changes, harmonics bloom, fade and hop. An aeolian harp mostly sings upper harmonics, rarely the fundamental. Nothing sounds without wind (or a pluck).

WIND up or thinner strings (faders down) → higher harmonics. **COUPLING** sets how strongly a harmonic holds on: low, harmonics swell as the wind passes them; high, a harmonic holds over a wide range of wind and then jumps.

Keys tune the harp (they do not play notes)

The last key is the root (raised to A1 at least); a key also plucks the strings it retunes, by velocity. Releasing a key changes nothing: the harp keeps its tuning. Retuning glides (GLIDE, FX page).

UNISON	every string on the root, a few cents apart: as an aeolian harp is strung
HARMONIC	the root's harmonic series 1–8 (just intonation)
FIFTHS	root, 5th, octave, 12th, 2 octaves, 19th, an octave and a 4th below
CHORD	the last eight keys (newest first), repeated an octave up

SOUND knobs

WIND	mean wind 0.5 – 15 m/s
GUSTS	gust size, 0 – 60 % of the wind
GUST TIME	how long a gust lasts, 0.2 – 10 s (LORENZ: how fast the weather turns)
COUPLING	lock-in strength (wake ↔ string)
RING	how long a string rings, 0.3 – 18 s
BRIGHT	how long the high harmonics ring, from 2 % of RING to all of it
AIR	wind noise on the strings
BODY	the sound box: dry strings → resonant box

Weather (Play / Stop / Record)

STEADY: the wind as set. **GUSTY**: random gusts (size GUSTS, length GUST TIME). **LORENZ**: the wind follows the Lorenz system (a model of convection): it swings between two moods and never repeats. Each string also sees a little turbulence of its own, so they never lock alike.

Pads

Top = presets 0–7. **Bottom**, while held; strike velocity, then pressure, sets the amount:

GUST	more wind, up to ×2.5
CALM	the wind dies down to nothing
STORM	bigger gusts, wind ×1.6
PLUCK	pluck every string (velocity)
MUTE	a hand on the strings
SLACK	strings down, up to 2 st
THROW / SWELL	delay throw / reverb swell

Wheels and transport

MOD: more wind (up to ×2). Pitch bend: string tension ±2 st. HOLD WIND (Quantise) freezes the wind where it is. PLUCK ALL (Click). DICE (Capture MIDI) rolls tuning, weather, SOUND page and gauges.

The model

Each string: a digital waveguide, excited at a point; its loop tuned exactly (to 0.06 cent in the tests). Each wake: a van der Pol oscillator (Facchinetti, de Langre & Biolley 2004),

$$\ddot{q}+\varepsilon\Omega(q^2-1)\dot{q}+\Omega^2 q = K\,\Omega^2\,v/L, \quad \Omega = 2\pi\,0.2\,U/d,$$

driving the string with the lift $L\,q$, $L \propto U^2 d$; v is the string's velocity at that point. Lock-in shows hysteresis, as in vortex-induced vibration: rising wind holds a harmonic longer. Gusts: an Ornstein–Uhlenbeck process; LORENZ: x of Lorenz-63. Code: `lib/aeolian_harp.h`.

Signal path and load

8 strings → sound box → level control (slow; the wind swings 40 dB) → stereo delay → Elements reverb → VOL. DSP 75.5 %, constant (no voices to add); patch RAM 21 of 44 KB.

Presets: top pads = 0–7; NEXT PRESET (fader button 9) steps; RESET (Undo) = 0

(chosen from the parameter meanings, not yet tuned by ear)

0 breeze	UNISON on A2, gusty 2.5 m/s, strings 0.4 – 2.2 mm: the 2nd to 5th harmonics take turns. The power-on sound.
1 still air	HARMONIC series on D2, steady 1.4 m/s, rings 12 s: a slow, held chord of harmonics.
2 gale	UNISON on E2, 7.3 m/s with big, short gusts (0.6 s), more noise: high harmonics, restless.
3 weather	FIFTHS on G2, the Lorenz wind: calm and stormy spells that never repeat.
4 fifths tower	FIFTHS on C2, gentle gusts, strong lock-in, long ring: steady, overlapping fifths.
5 glass strings	UNISON on A3, thin strings (0.25 – 0.6 mm), bright: the 7th to 10th harmonics, glassy.
6 low cables	HARMONIC on A1, thick strings (1.5 – 3 mm), 6.4 m/s, dark, long ring, slow glide.
7 chord field	CHORD: play a few keys, the last eight stay strung; Lorenz wind.

A preset sets tuning, weather, root, both knob pages and the eight diameters; the volume stays.

Sound design moves

Harmonic glissando: STEADY, COUPLING 3/4, sweep WIND (or the mod wheel) slowly: harmonics take over one by one; sweep back down and they hold longer (hysteresis). **One string**: switch the others off (fader buttons) and play WIND against its diameter. **Gust gestures**: GUST and CALM pads with pressure. **Bowed metal**: AIR 0, RING full, BRIGHT 3/4, COUPLING full. **A chord in the wind**: CHORD, play it, then let the weather run.

Where the values fall (knob turned . . .)

	0	1/4	1/2	3/4	full
WIND (m/s)	0.5	1.2	2.8	6.4	15
GUSTS (of the wind)	0	15 %	30 %	45 %	60 %
GUST TIME (s)	0.2	0.54	1.4	3.7	10
COUPLING K	0.001	0.0024	0.0056	0.013	0.03
RING (s)	0.3	0.84	2.4	6.4	18
BRIGHT (ring at 4 kHz, of RING)	2 %	5 %	14 %	37 %	100 %
diameter (fader, mm)	0.25	0.47	0.87	1.6	3
GLIDE (s)	0	0.25	1.0	2.2	4

The Strouhal table: shedding frequency $0.2\,U/d$ (Hz)

$d \setminus U$	1 m/s	3 m/s	5 m/s	10 m/s	15 m/s
0.25 mm	800	2400	4000	8000	(12000)
0.5 mm	400	1200	2000	4000	6000
1 mm	200	600	1000	2000	3000
2 mm	100	300	500	1000	1500
3 mm	67	200	333	667	1000

Divide by the string's pitch for the harmonic it sings (A2 = 110 Hz: 1 mm at 3 m/s → the 5th to 6th). Above 12 kHz the wind no longer drives a string.

Something seems wrong?

Silence: WIND at the bottom, CALM held, all strings off, or VOL down. A string sings only where its shedding meets a harmonic: move WIND a little. **Only hiss**: AIR high and the wind too strong for the strings (shedding above their harmonics): thicker strings or less wind. **Frozen**: HOLD WIND latched (Quantise). **Lost**: RESET (Undo).